WEEK 2

TRIGGERS:

```
mysql> USE transport_db;
Database changed
mysql> CREATE TABLE driver(
    -> driver_id INT PRIMARY KEY,
    -> driver_name VARCHAR(50),
    -> otp INT
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER before_driver_insert
    -> BEFORE INSERT ON driver
    -> FOR EACH ROW
    -> BEGIN
    -> F NEW.otp < 100000 OR NEW.otp > 999999 THEN
    -> F NEW.otp < 100000 OR NEW.otp > 999999 THEN;
    -> ;
    -> //
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'NEW.otp < 100000 OR NEW.otp > 999999 THEN
F NEW.otp < 100000 OR NEW.otp > 999999' at line 5
mysql> DELIMITER //
mysql> CREATE TRIGGER before_driver_insert
    -> BEFORE INSERT ON driver
    -> FOR EACH ROW
    -> BEGIN
    -> IF NEW.otp < 100000 OR NEW.otp > 999999 THEN
    -> SIGNAL SQLSTATE '45000'
    ->  SET MESSAGE_TEXT = 'OTP must be a 6-digit number';
    -> END IF;
    -> END;
```

```
Query OK, 0 rows affected (0.13 sec)

mysql> INSERT INTO driver VALUES (1, 'Ravi', 12345);
    -> INSERT INTO driver VALUES (2, 'Arun', 123456);
    -> //
ERROR 1644 (45000): OTP must be a 6-digit number
mysql> INSERT INTO driver VALUES (2, 'Arun', 123456);
    -> //
Query OK, 1 row affected (0.03 sec)

mysql> CREATE TABLE orders (
    -> order_id INT PRIMARY KEY,
    -> total_price DECIMAL(10,2) DEFAULT 0
    -> );
    -> //
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE order_items (
    -> item_id INT PRIMARY KEY AUTO_INCREMENT,
    -> order_id INT,
    -> dish_name VARCHAR(50),
    -> price DECIMAL(10,2)
    -> );
    -> //
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER after_item_insert
    -> AFTER INSERT ON order_items
    -> FOR EACH ROW
    -> BEGIN
    ->  UPDATE orders
```

```
    -> //
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO orders VALUES (1, 0);
    -> INSERT INTO order_items (order_id, dish_name, price)
    -> VALUES (1, 'Pizza', 250);
    -> SELECT * FROM orders;
    -> //
Query OK, 1 row affected (0.02 sec)

Query OK, 1 row affected (0.08 sec)

+----------+-------------+
| order_id | total_price |
+----------+-------------+
|        1 |      250.00 |
+----------+-------------+
1 row in set (0.09 sec)

mysql> CREATE TABLE users (
    -> user_id INT PRIMARY KEY,
    -> username VARCHAR(50),
    -> created_at TIMESTAMP
    -> );
    -> //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER before_user_insert
    -> BEFORE INSERT ON users
    -> FOR EACH ROW
    -> BEGIN
    -> SET NEW.created_at = CURRENT_TIMESTAMP;
```

```
mysql> INSERT INTO users VALUES (1, 'Divij', NULL);
    -> SELECT * FROM users;
    -> //
Query OK, 1 row affected (0.03 sec)

+---------+----------+---------------------+
| user_id | username | created_at          |
+---------+----------+---------------------+
|       1 | Divij    | 2026-02-04 09:29:03 |
+---------+----------+---------------------+
1 row in set (0.03 sec)

mysql> CREATE TABLE employee (
    -> emp_id INT PRIMARY KEY,
    -> emp_name VARCHAR(50),
    -> salary DECIMAL(10,2)
    -> );
    -> //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER before_employee_insert
    -> BEFORE INSERT ON employee
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary cannot be negative';
    ->     END IF;
    -> END;
    -> //
Query OK, 0 rows affected (0.02 sec)
```

```
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_employee_update
    -> BEFORE UPDATE ON employee
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary cannot be neg
    ->     END IF;
    -> END;
    -> //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> INSERT INTO employee VALUES (1, 'Amit', -5000);
ERROR 1644 (45000): Salary cannot be negative
mysql> INSERT INTO employee VALUES (2, 'Neha', 30000);
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE employee_audit (
    ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_id INT,
    ->     old_emp_name VARCHAR(50),
    ->     old_salary DECIMAL(10,2),
    ->     changed_at TIMESTAMP
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> DELIMITER //
mysql>
```

```
    ->     (emp_id, old_emp_name, old_salary, changed_at)
    ->     VALUES
    ->     (OLD.emp_id, OLD.emp_name, OLD.salary, NOW());
    -> END;
    -> //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> UPDATE employee
    -> SET salary = 45000
    -> WHERE emp_id = 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM employee_audit;
+----------+--------+--------------+------------+---------------------+
| audit_id | emp_id | old_emp_name | old_salary | changed_at          |
+----------+--------+--------------+------------+---------------------+
|        1 |      2 | Neha         |   30000.00 | 2026-02-04 09:37:25 |
+----------+--------+--------------+------------+---------------------+
1 row in set (0.00 sec)
```

Working with Date:

```
mysql> SELECT DATEDIFF('2026-02-10', '2026-02-01') AS days_difference;
+-----------------+
| days_difference |
+-----------------+
|               9 |
+-----------------+
1 row in set (0.03 sec)

mysql> SELECT DATE_ADD('2026-02-01', INTERVAL 30 DAY) AS expiry_date;
+-------------+
| expiry_date |
+-------------+
| 2026-03-03  |
+-------------+
1 row in set (0.02 sec)

mysql> SELECT DATE_SUB(CURDATE(), INTERVAL 7 DAY) AS date_before_7_days;
+--------------------+
| date_before_7_days |
+--------------------+
| 2026-01-28         |
+--------------------+
1 row in set (0.00 sec)

mysql> CREATE TABLE records (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     created_at DATE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE records (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     created_at DATE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT *
    -> FROM records
    -> WHERE DAYOFWEEK(created_at) IN (1, 7);
Empty set (0.03 sec)

mysql> SELECT
    ->     MONTHNAME(
    ->         STR_TO_DATE('15-02-2026', '%d-%m-%Y')
    ->     ) AS month_name;
+------------+
| month_name |
+------------+
| February   |
+------------+
1 row in set (0.02 sec)
```

Indexes and Views:

```
mysql> CREATE TABLE users1 (
    -> user_id INT PRIMARY KEY,
    ->     username VARCHAR(50),
    ->     email VARCHAR(100),
    ->     aadhar BIGINT,
    ->     status VARCHAR(20),
    ->     created_at DATETIME
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE INDEX idx_email
    -> ON users1(email);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM users1 WHERE email = 'abc@gmail.com';
Empty set (0.03 sec)

mysql> CREATE UNIQUE INDEX idx_unique_username
    -> ON users1(username);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO users1 VALUES (1, 'divij', 'a@gmail.com', 1234, 'active', NOW());
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO users1 VALUES (2, 'divij', 'b@gmail.com', 5678, 'active', NOW());
ERROR 1062 (23000): Duplicate entry 'divij' for key 'users1.idx_unique_username'
mysql> INSERT INTO users1 VALUES (2, 'dj', 'b@gmail.com', 5678, 'active', NOW());
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SHOW INDEX FROM users1;
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name           | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index
_type | Comment | Index_comment | Visible | Expression |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
| users1 |          0 | PRIMARY            |            1 | user_id     | A         |           2 |     NULL |   NULL |      | BTREE
      |         |               | YES     | NULL       |
| users1 |          0 | idx_unique_username |            1 | username    | A         |           2 |     NULL |   NULL | YES  | BTREE
      |         |               | YES     | NULL       |
| users1 |          1 | idx_email          |            1 | email       | A         |           2 |     NULL |   NULL | YES  | BTREE
      |         |               | YES     | NULL       |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
3 rows in set (0.08 sec)

mysql> SHOW INDEX FROM users1;
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name           | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index
_type | Comment | Index_comment | Visible | Expression |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
| users1 |          0 | PRIMARY            |            1 | user_id     | A         |           2 |     NULL |   NULL |      | BTREE
      |         |               | YES     | NULL       |
| users1 |          0 | idx_unique_username |            1 | username    | A         |           2 |     NULL |   NULL | YES  | BTREE
      |         |               | YES     | NULL       |
| users1 |          1 | idx_email          |            1 | email       | A         |           2 |     NULL |   NULL | YES  | BTREE
      |         |               | YES     | NULL       |
+--------+------------+--------------------+--------------+-------------+-----------+-------------+----------+--------+------+------
------+---------+---------------+---------+------------+
```

```
mysql> SELECT * FROM users1 WHERE aadhar = 123456789012;
Empty set (0.00 sec)

mysql> CREATE INDEX idx_aadhar
    -> ON users(aadhar);
ERROR 1072 (42000): Key column 'aadhar' doesn't exist in table
mysql> CREATE INDEX idx_aadhar
    -> ON users1(aadhar);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_status_created
    -> ON users1(status, created_at);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT *
    -> FROM users1
    -> WHERE status = 'active'
    -> ORDER BY created_at;
+---------+----------+-------------+--------+--------+---------------------+
| user_id | username | email       | aadhar | status | created_at          |
+---------+----------+-------------+--------+--------+---------------------+
|       1 | divij    | a@gmail.com |   1234 | active | 2026-02-04 09:41:32 |
|       2 | dj       | b@gmail.com |   5678 | active | 2026-02-04 09:42:12 |
+---------+----------+-------------+--------+--------+---------------------+
2 rows in set (0.00 sec)
```

SQL Built-in Function

```
mysql> SELECT
    ->     CONCAT(SUBSTRING_INDEX(username, '@', 1), '123') AS default_password
    -> FROM users1;
+------------------+
| default_password |
+------------------+
| divij123         |
| dj123            |
+------------------+
2 rows in set (0.01 sec)
```

```
mysql> SELECT
    ->     username,
    ->     CONCAT(SUBSTRING_INDEX(username, '@', 1), '123') AS default_password
    -> FROM users1;
+----------+------------------+
| username | default_password |
+----------+------------------+
| divij    | divij123         |
| dj       | dj123            |
+----------+------------------+
2 rows in set (0.00 sec)

mysql> SELECT
    ->     aadhar,
    ->     REGEXP_REPLACE(aadhar, '[^0-9]', '') AS valid_aadhar
    -> FROM users1;
+--------+--------------+
| aadhar | valid_aadhar |
+--------+--------------+
|   1234 | 1234         |
|   5678 | 5678         |
+--------+--------------+
2 rows in set (0.04 sec)
```

```
    ->     CONCAT(SUBSTRING_INDEX(username, '@', 1), '123') AS default_password
    -> FROM users1;
----------+------------------+
 username | default_password |
----------+------------------+
 divij    | divij123         |
 dj       | dj123            |
----------+------------------+
2 rows in set (0.00 sec)

ysql> SELECT
    ->     aadhar,
    ->     REGEXP_REPLACE(aadhar, '[^0-9]', '') AS valid_aadhar
    -> FROM users1;
--------+--------------+
 aadhar | valid_aadhar |
--------+--------------+
   1234 | 1234         |
   5678 | 5678         |
--------+--------------+
2 rows in set (0.04 sec)

ysql> SELECT
    ->     username,
    ->     CONCAT(
    ->         UPPER(SUBSTRING(username, 1, 1)),
    ->         LOWER(SUBSTRING(username, 2))
    ->     ) AS normalized_name
    -> FROM users1;
----------+-----------------+
 username | normalized_name |
----------+-----------------+
```

```
    ->     ) AS normalized_name
    -> FROM users1;
+----------+-----------------+
| username | normalized_name |
+----------+-----------------+
| divij    | Divij           |
| dj       | Dj              |
+----------+-----------------+
2 rows in set (0.00 sec)

mysql> SELECT
    ->     COALESCE(email, username) AS preferred_contact
    -> FROM users1;
+-------------------+
| preferred_contact |
+-------------------+
| a@gmail.com       |
| b@gmail.com       |
+-------------------+
2 rows in set (0.00 sec)

mysql> SELECT
    ->     dept,
    ->     AVG(TIMESTAMPDIFF(YEAR, joining_date, CURDATE())) AS avg_experience
    -> FROM employee
    -> GROUP BY dept;
ERROR 1054 (42S22): Unknown column 'dept' in 'field list'
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_net_salary(gross_salary DECIMAL(10,2))
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
```

User-Defined Functions:

```
mysql> CREATE FUNCTION calculate_net_salary(gross_salary DECIMAL(10,2))
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN gross_salary - (gross_salary * 0.10);
    -> END;
    -> //
Query OK, 0 rows affected (0.06 sec)

mysql> DELIMITER ;
mysql> SELECT calculate_net_salary(50000) AS net_salary;
+------------+
| net_salary |
+------------+
|   45000.00 |
+------------+
1 row in set (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION check_user_activity(last_login DATE)
    -> RETURNS VARCHAR(10)
    -> DETERMINISTIC
    -> BEGIN
    ->     IF DATEDIFF(CURDATE(), last_login) <= 30 THEN
    ->         RETURN 'Active';
    ->     ELSE
    ->         RETURN 'Inactive';
    ->     END IF;
    -> END;
    -> //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER ;
mysql> SELECT check_user_activity('2026-01-20') AS user_status;
+-------------+
| user_status |
+-------------+
| Active      |
+-------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION calculate_tax(salary DECIMAL(10,2))
    -> RETURNS DECIMAL(10,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE tax DECIMAL(10,2);
    ->
    ->     IF salary <= 300000 THEN
    ->         SET tax = 0;
    ->     ELSEIF salary <= 600000 THEN
    ->         SET tax = salary * 0.10;
    ->     ELSEIF salary <= 1000000 THEN
    ->         SET tax = salary * 0.20;
    ->     ELSE
    ->         SET tax = salary * 0.30;
    ->     END IF;
    ->
    ->     RETURN tax;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
```

```
mysql> DELIMITER ;
mysql> SELECT calculate_tax(750000) AS tax_amount;
+------------+
| tax_amount |
+------------+
|  150000.00 |
+------------+
1 row in set (0.00 sec)
```